

Ehsan Sharafian

 ehsan.sharafian@maine.edu

 (207) 631-6979

 LinkedIn

 GitHub

 ehsansharafian.com

 Google Scholar

PROFESSIONAL SUMMARY

Mechanical Engineering PhD candidate specializing in intelligent wearable systems for human movement analysis and digital health. Experienced in building end-to-end solutions that integrate IMU-based sensing, embedded hardware, smartphone applications, machine learning, and haptic feedback for gait analysis and rehabilitation. Skilled in Python, MATLAB, Android Studio/Java, and PyTorch, with six peer-reviewed journal publications.

EDUCATION

- Ph.D. in Mechanical Engineering | University of Maine, Orono, ME

Expected Dec 2027

GPA: 4.00/4.00 | Dissertation: Smart Real-Time Gait Training System
- M.S. in Mechanical Engineering | Amirkabir University of Technology (Tehran Polytechnic)

Feb 2020

GPA: 3.65/4.00 | Thesis: Trajectory optimization of a robot arm in joint and Cartesian spaces
- B.S. in Mechanical Engineering | University of Tehran

Jul 2017

TECHNICAL SKILLS

Programming & Data Analysis	Python, Java, C/C++, NumPy, Pandas, scikit-learn, PyTorch, SPSS
Mobile & Software	MATLAB, Simulink, TwinCAT, Android Studio, Unity, Git
Wearable & Embedded Systems	IMU sensors, Arduino IDE, SparkFun modules, vibrotactile motors, PCB modules
Biomechanics & Signal Processing	Gait analysis, human activity recognition, sensor fusion, feature extraction, OpenSim
Robotics	Surgical robotics, parallel manipulators, inverse kinematics, dynamics, control systems
CAD	SolidWorks

RESEARCH & ENGINEERING EXPERIENCE

- Graduate Research Assistant | Biorobotics and Biomechanics Lab, University of Maine

Sep 2023 – Present

Orono, ME
- Led the development of real-time wearable sensing systems and Android-based mobile applications for gait analysis, activity recognition, fall-risk assessment, and home-based rehabilitation using IMUs, haptic feedback, and smartphone-deployed feedback platforms.
 - Built a smartphone-deployed gait assessment system using a single foot-mounted IMU to reconstruct foot-height trajectories, achieving **96.08% locomotion classification accuracy**, **<1.1 cm cumulative height error**, and **112 ms latency**.
 - Engineered a lightweight human activity recognition system using two IMUs, biomechanical feature extraction, and supervised classifiers including SVM and Naive Bayes, achieving **99.1% accuracy** for normal locomotion and **93.1% accuracy** for atypical stair-negotiation strategies in older adults.
 - Architected a home-based haptic feedback gait-training system integrating IMUs, vibrotactile motors, custom Android applications, Arduino/SparkFun modules, and PCB circuitry for adults aged 65+.
 - Demonstrated up to **30% improvement** in stride length and walking speed during a single-session haptic feedback study, with minimal cognitive demand.
 - Led IRB-approved human-subject validation studies with **85+ participants** across gait analysis, activity recognition, and haptic feedback protocols.

- Developed real-time algorithms and data pipelines in Android Studio/Java, Python, and MATLAB, combining signal processing, sensor fusion, biomechanical modeling, and machine learning for mobile health applications.

Mechanical and Control Systems Engineer | [Sina Robotics and Medical Innovators Co.](#)

Tehran, Iran

Dec 2022 – Sep 2023

- Developed PLC-based control logic for a telesurgical laparoscopic robotic platform using Beckhoff TwinCAT and IEC 61131-3 Structured Text, with MATLAB/Simulink validation of inverse kinematics and initialization workflows.
- Built a Python/Kivy graphical user interface on Raspberry Pi for surgeon-facing robot control, system guidance, warnings, and improved human-robot interaction during testing and demonstrations.
- Coordinated hands-on validation sessions with **6 experienced surgeons**, using training, troubleshooting, and usability feedback to refine surgeon-facing robotic system workflows.

Laboratory Research Assistant | [New Technology Research Center, Amirkabir University of Technology](#)

Tehran, Iran

Nov 2020 – Nov 2022

- Designed and assembled robotics and mechatronics setups for student research projects, integrating hardware, CAD, MATLAB workflows, and technical documentation.
- Mentored 6 undergraduate students across 5 research teams, guiding equipment use, troubleshooting, documentation, and project execution.
- Coordinated technical preparation for the 9th International Conference on Robotics and Mechatronics, including lab demonstrations, equipment readiness, and presentation logistics.

SELECTED HONORS

Flagship Fellowship Award, University of Maine

Graduate Student Government Award for conference proposal

Sep 2024 – Sep 2025

Mar 2025

LEADERSHIP & SERVICE

Graduate Senator, Department of Mechanical Engineering, University of Maine

President, Iranian Graduate Student Association, University of Maine

Sep 2025 – Present

Sep 2024 – Apr 2026

PUBLICATIONS

6 peer-reviewed journal articles (3 first-author) in wearable sensing, biomechanics, rehabilitation engineering, and robotics.

- **Sensors** (2026) – Real-time foot-height estimation and activity classification
- **IEEE Access** (2025) – Lightweight IMU-based activity recognition for older adults
- **IEEE TNSRE** (2025) – Haptic feedback effects on gait and cognitive load
- **Journal of Biomechanics** (2025) – Thigh extension and leg coordination
- **International Journal of Robotics and Automation** (2023) – Trajectory optimization of a robot arm
- **Robotica** (2022) – Screw theory-based constraint wrench analysis